



CANDIDATE
NAME

CLASS

2T

INDEX
NUMBER

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BIOLOGY

9648/02
23rd August 2016
2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are two sections in this paper.

Section A]

Answer all questions

Section B]

Answer all questions. Answer each part on a **separate** piece of paper.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

For Examiner's Use	
Section A	80
1 [10]	
2 [10]	
3 [10]	
4 [10]	
5 [10]	
6 [10]	
7 [10]	
8 [10]	
Section B	20
9a/10a	
9b/10b	
9c/10c	
TOTAL	45

Section A

Answer **all** questions in this section.

- 1 The figure below shows three structural components of the membrane system in a pancreatic cell.

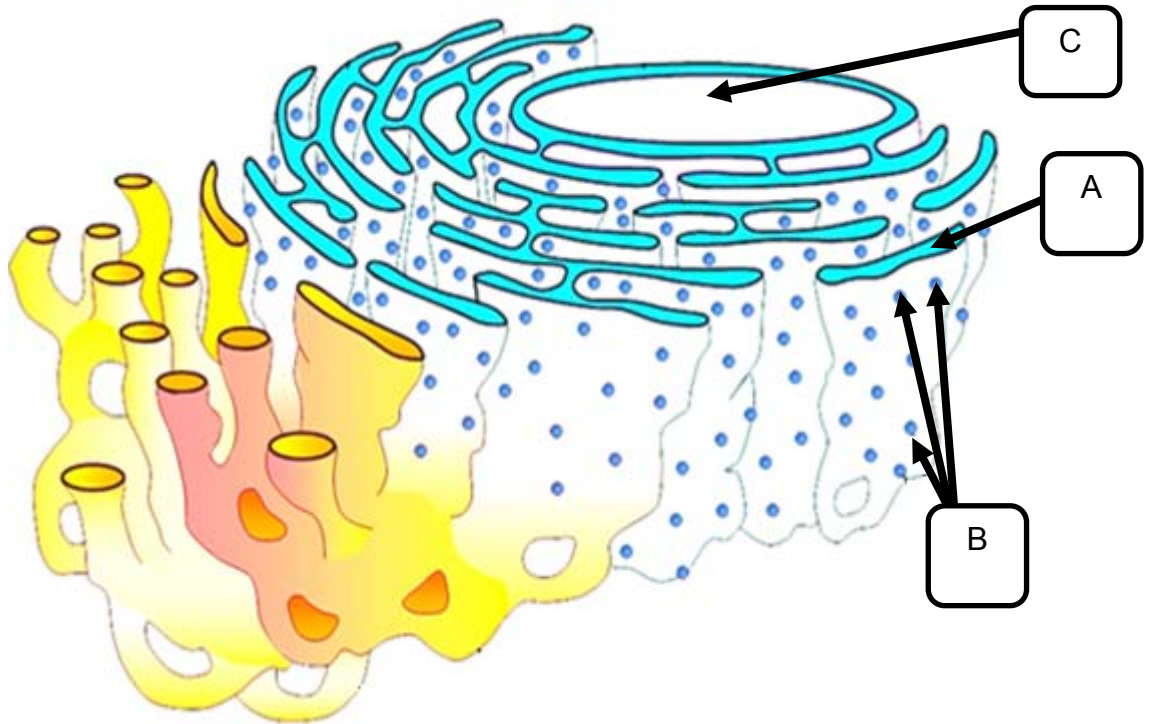


Fig. 1.1

- (a) State the name and function of the structures in Fig. 1.1.

Structure A

Name:

Function: [1]

1. Cisternae of rough endoplasmic reticulum / rER, involved in structural folding of primary polypeptide.

[L1]

Structure B

Name:

Function: [1]

2. Bound ribosomes of rough ER, involved in the synthesis of polypeptide chain which enters the rER.

[L1]

- (b) State what is **structure C** and explain how **structure A** remains in close proximity to **C**.

.....

SC: Structure C / How Structure A remains close to structure C

OR: Nucleus / continuous membrane with nucleus

1. Nucleus.
2. The nuclear envelope / membrane / double membrane of the nucleus is continuous with the membrane system of the rough ER.

[L2]

Fig. 1.2 shows a hormone essential for regulation of high blood glucose levels.

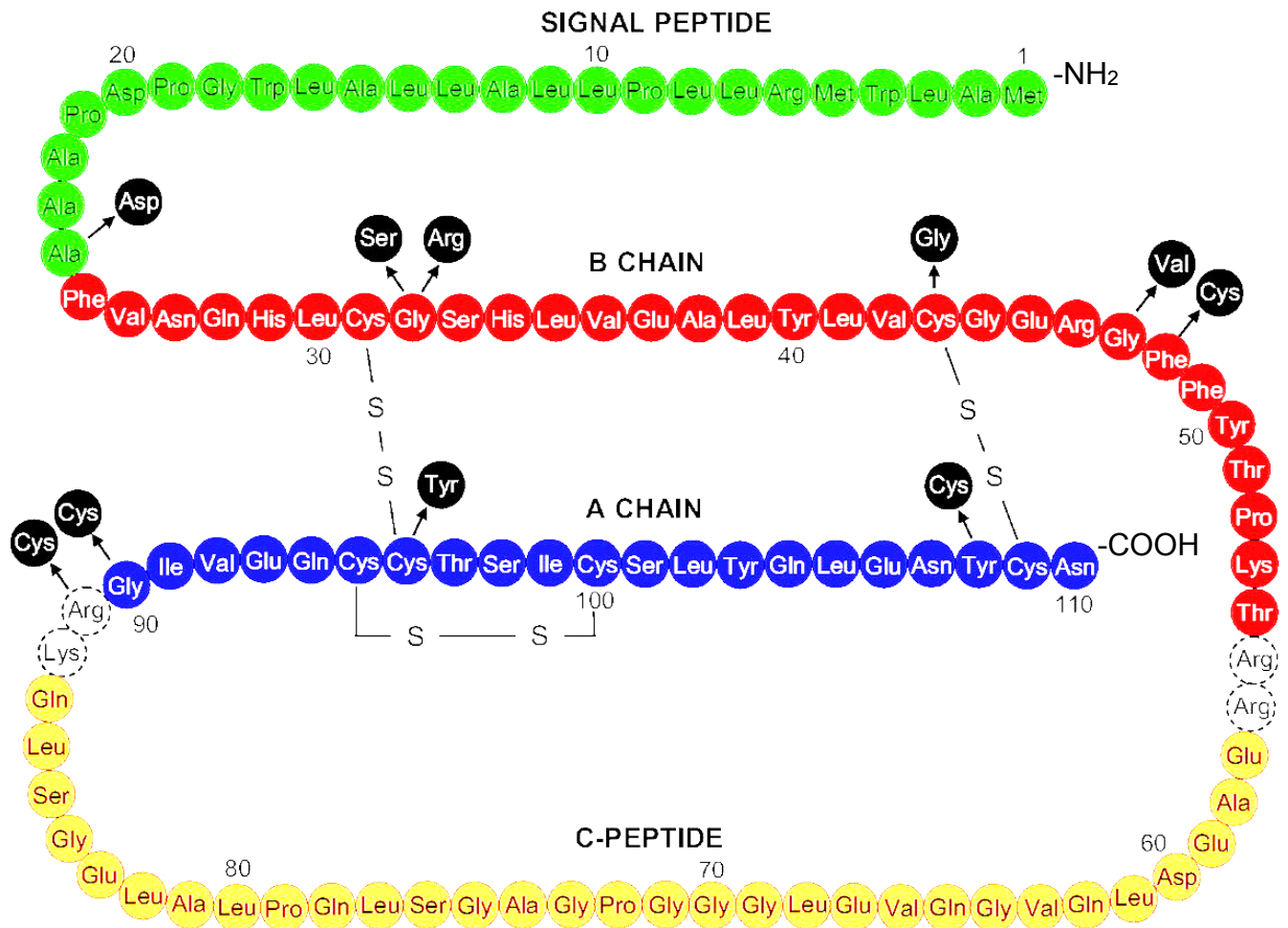


Fig. 1.2 Unit sequences of wild type (WT) and the mutant hormone. The mutations are noted in black circles together with location in the B (25 to 5) or A chain (90-110). The dashed circles indicate the basic residues that are activation cleavage sites.

- (c) With reference to Fig. 1.2, draw out the bond (ensure correct orientation) formed between units 109 and 110 of the A-chain.

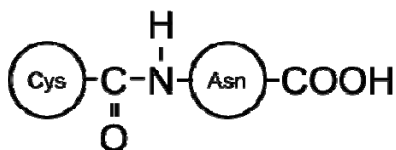


[2]

SC: Draw / Bond / orientation

OR: Peptide bond / COOH on right same as in aa 110

1. Peptide bond
2. Correct orientation



[L2]

- (d) With reference to Fig 1.2, suggest how amino acid substitutions between 90 and 109 could occur and what would be the corresponding effect on the resultant mutant hormone.

.....
.....
.....
.....[4]

SC: how aa substitution occur / effect on hormone

OR: base pair substitutions on template strand

- change amino acid Example 96 Cytosine / Cys to Tyrosine / Tys
- Affects the disulphide bond between 31 Cys and 96 Cys
- Changing the quaternary structure [chain A and B >2 molecules]
- Change in overall configuration of hormone

1. Base pair substitution on the non coding strand of DNA.

Any pair 2 & 3 or 4 & 5

2. results in a change in codon for a specific amino acid 96 Cys to Tys.
3. resulting in the loss of a disulphide bond between 96 Cys and 31 Cys.

Any 1 of the following:

4. results in a change in codon for a specific amino acid 108 Tyr to Cys.
5. resulting in the gain of a disulphide bond between 43 Cys and 108 Cys.

6. changing the quaternary structure of the hormone and its overall specific configuration.

(Max 4)

[L3]

[Total: 10]

- 2 The figure shows the molecular configuration of chymotrypsin. Chymotrypsin is one of the major proteases in the human digestive tract, in which its role is to hydrolyse large protein molecules into smaller peptides that are then further processed by peptidases. Fig. 2.1a shows a blown up representation of a portion of chymotrypsin shown in Fig. 2.1b.

Fig. 2.1a

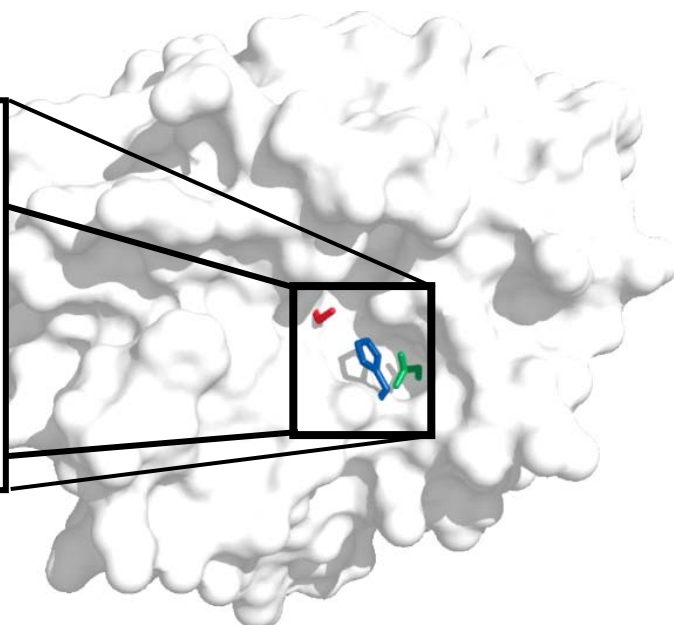
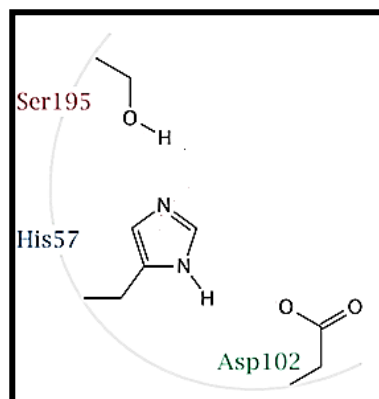


Fig. 2.1b

- (a) Using the 'induced-fit hypothesis', explain the mechanism in which chymotrypsin carries out its function.

.....

[2]

SC: Induced fit / mechanism / chymotrypsin / function.

OR: 1. In the induced-fit hypothesis, catalytic R groups of the active site come into correct orientation and bind.
 2. The binding causes conformational change that fits the enzyme more closely with the substrate and in so doing a strain on the substrate bond to be broken.

1. In the 'induce fit' hypothesis, catalytic R groups of the active site come into correct orientation and bind to the protein.
2. The bind causes conformational change that fits the enzyme more closely with the substrate.
3. and in so doing causes a strain in the structural bond lowering the activation energy of the reaction.

(Max 2)

[L2]

Fig. 2.2 shows the energy exchange in a chemical reaction without the enzyme present.

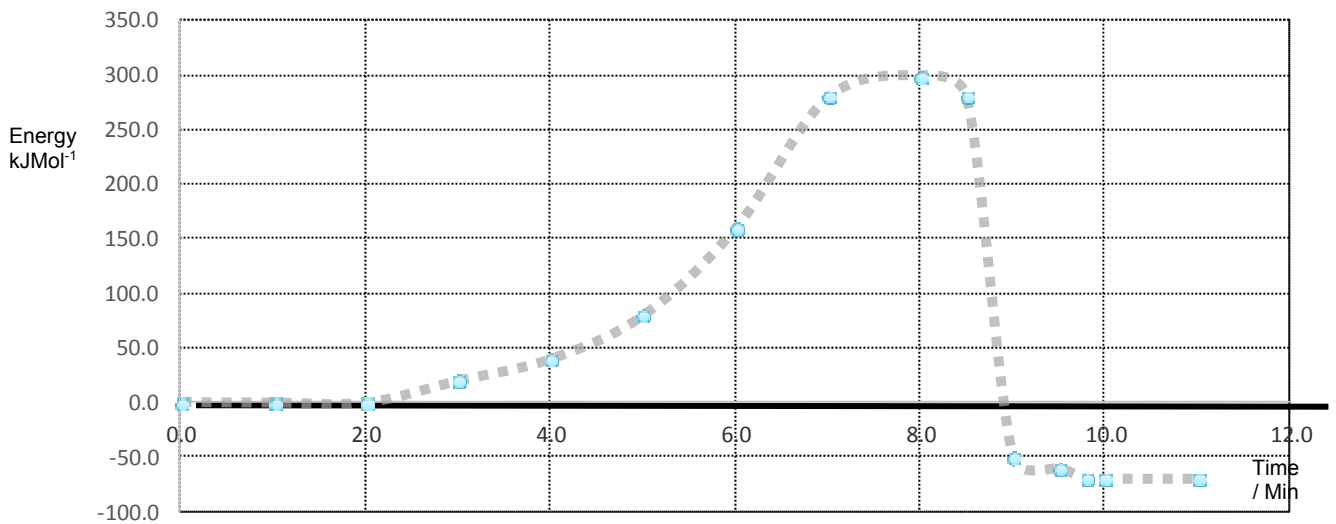
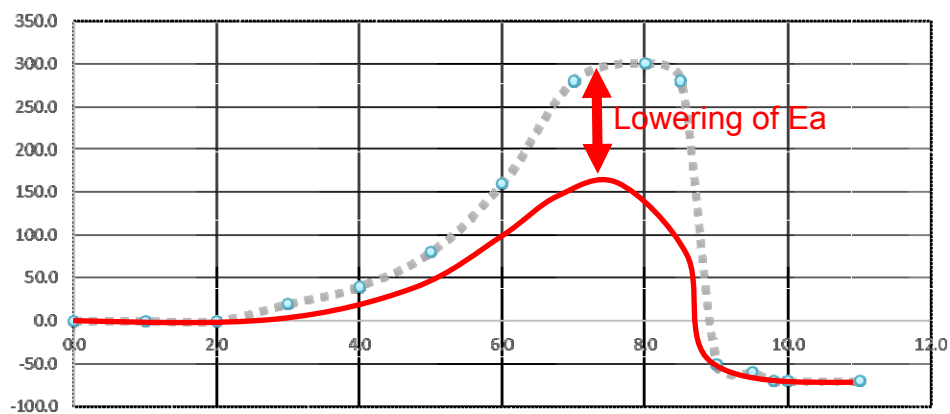


Fig. 2.2

- (b) On the graph above draw out the plot tracing the effect of enzyme action and label the effect of enzyme on activation energy. [2]



Correct drawing of the enzyme reaction graph [1 mark]
Lowering of Ea. [1 mark]

[L2]

Fig. 2.3 shows the effects of substrate on enzyme reaction.

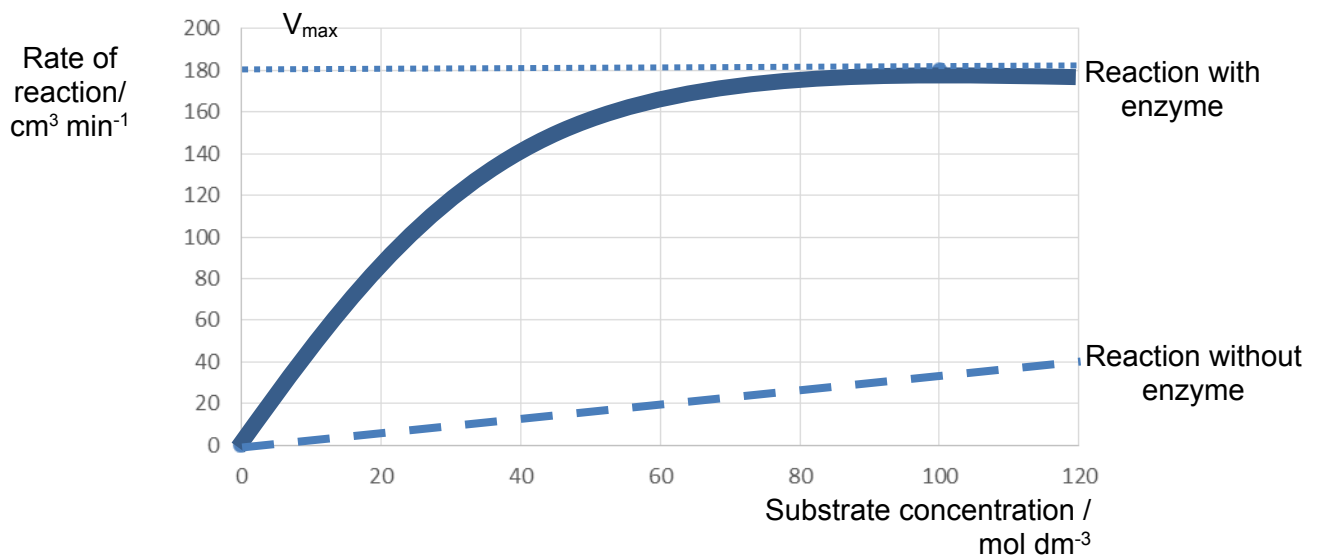


Fig. 2.3

- (c) Explain what can be inferred from the graph in Fig. 2.3, with reference to substrate concentration and limiting factors.

.....
[2]

SC: inferred from graph / ref Sub con. & limiting factors..

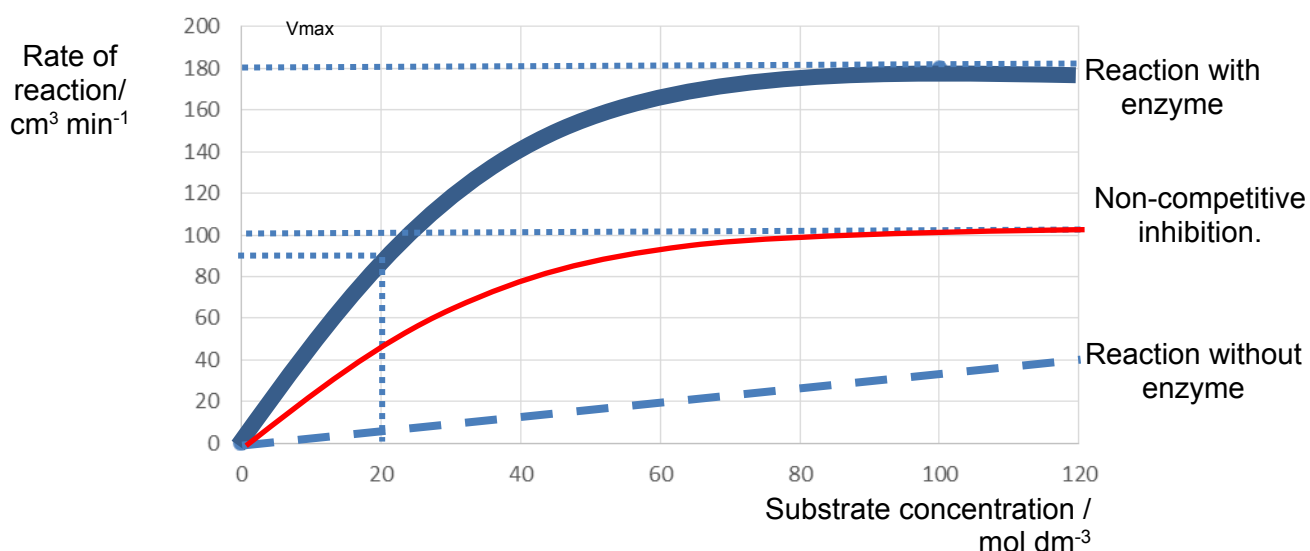
OR: 1. DRUM substrate con below 80 mol dm^{-3} / limiting factor / adding more rate of reaction increases.
 2. From 80 mol dm^{-3} onward, substrate concentration no longer is the limiting factor

1. From 0 to 80 mol dm^{-3} substrate is the limiting factor since upon adding more substrate sees the rate of reaction increasing.
 2. From 80 mol dm^{-3} onward, substrate concentration is no longer the limiting factor.
 3. Citation of Data.
 (Max 2)

[L2]

- (d) On Fig. 2.3 draw a curve representing the effect of a non-competitive inhibitor on rate of reaction with no change in the affinity of the enzyme.

[2]



Non-competitive inhibitor graph plateaus and does not reach V_{max} . [1 mark]
 K_m values of Reaction with enzyme and that of the Non-competitive are the same. [1 mark]

[L2]

- (e) Account for why the Michaelis constant for both the non-competitive inhibitor and the reaction without inhibitor is the same.

.....
[2]

SC: Michaelis Const / both rxn SAME

OR: 1. Michaelis constant (K_m) / $\frac{1}{2} V_{max}$

2. Since binding of the inhibitor / a site other than the active site / allosteric site there is no change in affinity / K_m value remains the same.
1. Michaelis constant (K_m) is derived by measuring $\frac{1}{2}$ V_{max} of each respective graph..
2. Since binding of the inhibitor is at a site other than the active site / allosteric site there is no change in affinity of the active site of the enzyme to its substrate and K_m value remains the same.
3. The 3D configuration of the active site is changed with the binding of the inhibitor.
- (Max 2)

[L2]

[Total: 10]

- 3 Cells were transferred and grown in ^{15}N medium for many generations before they were transferred to ^{14}N medium again and allowed to divide.

DNA was extracted periodically from the culture and subjected to density gradient centrifugation using caesium chloride.

Fig. 3.1 shows the density gradient results across three generations.

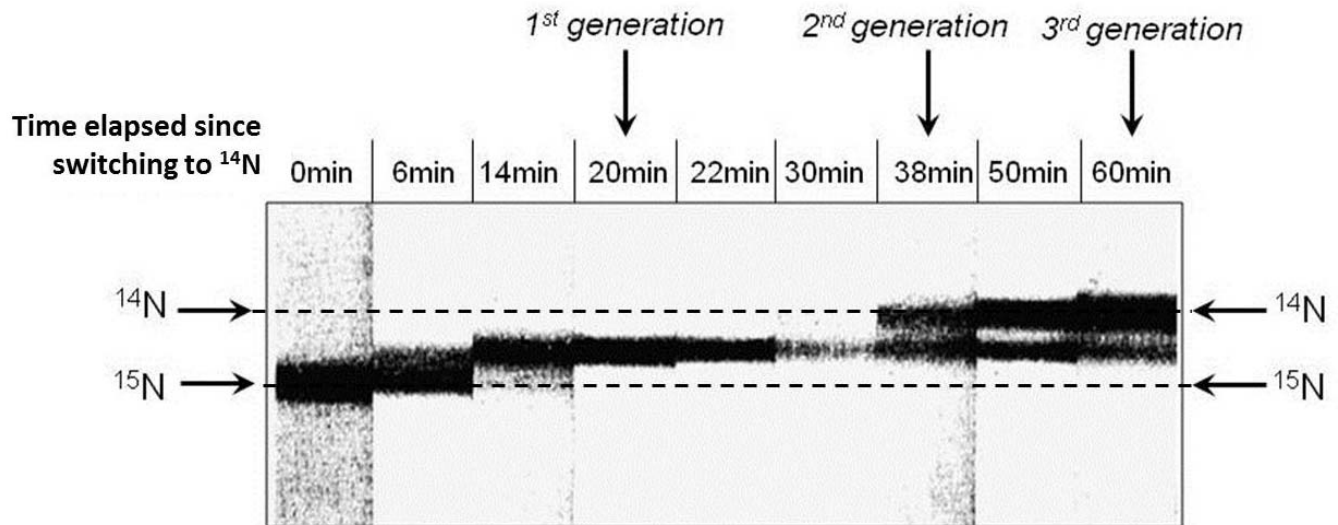


Fig. 3.1

- (a) With reference to Fig. 3.1, account for the model of DNA replication which these cells undergo.

.....

[3]

S: [Account](#) (CW), [with reference..Fig. 3.1](#), [model..DNA replication](#), [cells...undergo](#),

C:

- With reference Fig. 3.1 → Must cite information from Fig. 3.1 in answers
- There are 3 models of DNA replication: Semi-conservative, conservative, dispersive
- Model here is semi-conservative
 - 1 intermediate band at 1st generation
 - 2 intermediate band, 2 bands at 2nd generation, 2 bands at 3rd generation with intermediate band becoming thinner and light band becoming thicker.

ORE:

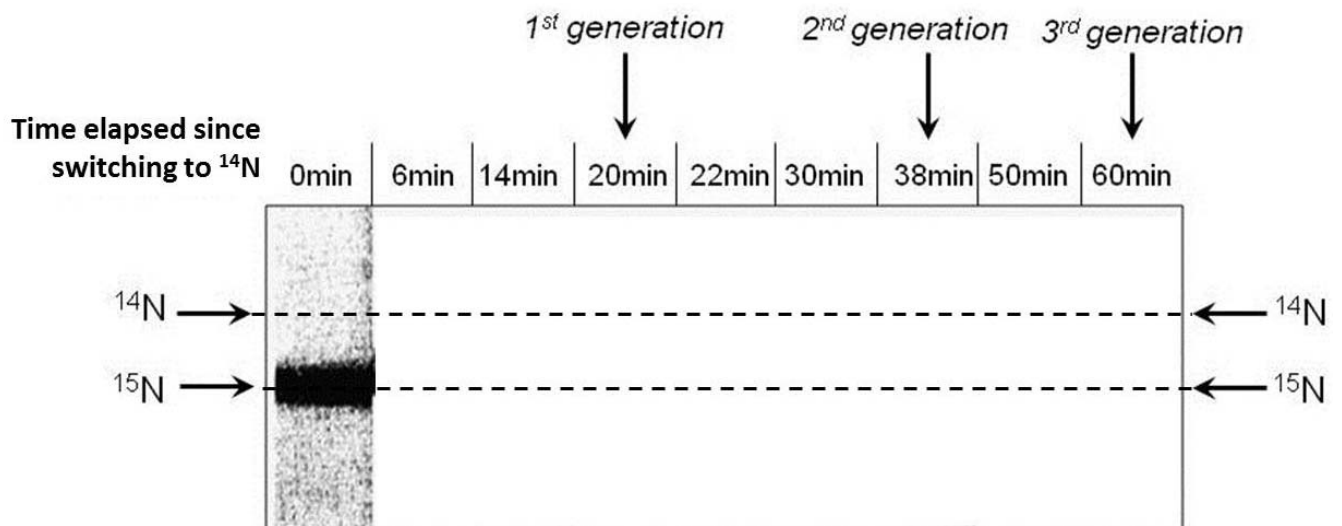
1. The model of DNA replication is semi-conservative.
2. At the first generation, there is only one $^{14}\text{N}/^{15}\text{N}$ / hybrid band, which suggests the parental strands, that contain ^{15}N separate to serve as template for the synthesis of the newly synthesised strand, which contains ^{14}N .
3. At the second generation, there is one $^{14}\text{N}/^{15}\text{N}$ / hybrid band and 1 light band which align with the semi-conservative model of replication as the former contains DNA with 1 strand containing ^{15}N and another containing ^{14}N , whereas the latter contains DNA with both strands containing ^{14}N .
4. For the third generation, there is still one $^{14}\text{N}/^{15}\text{N}$ / hybrid band and 1 $^{14}\text{N}/^{14}\text{N}$ band and the intermediate band becomes thinner whereas the light band becomes thicker due to more DNA molecules containing ^{14}N on both strands.

(Max 3)

[L2]

- (b) State another model of DNA replication not shown in Fig. 3.1 draw only its band patterns for the 1st, 2nd and 3rd generations in the Figure below.

Model of DNA replication:



[2]

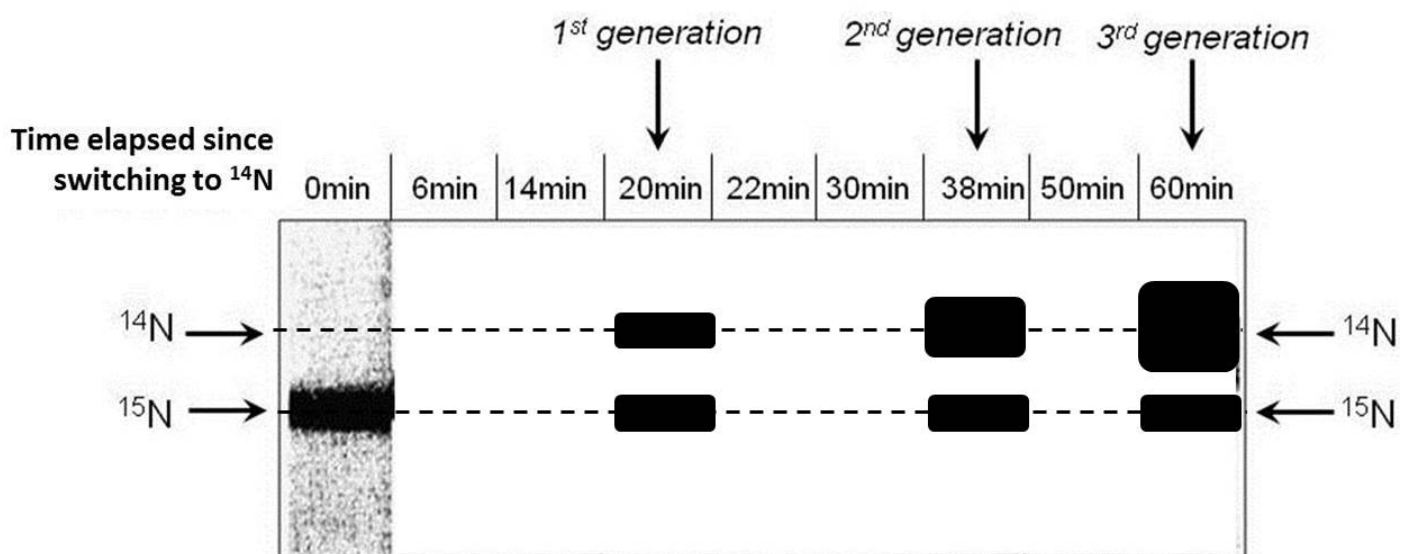
S: State....Draw (CW), model of DNA replication, not shown in Fig. 3.1, draw..band patterns..1st,2nd,3rd,

C:

- Model of DNA replication shown in Fig. 3.1: Semi-conservative
- Other models: Conservative & Dispersive

ORE:

- Model of DNA replication: Conservative replication



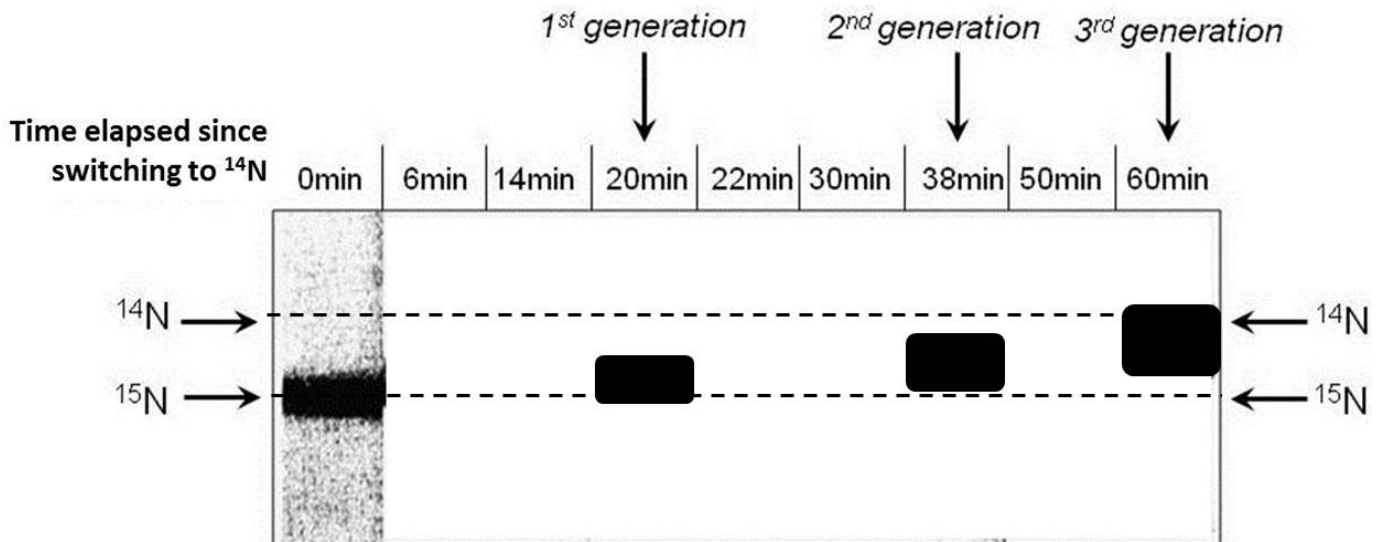
Correct positioning of the bands. [1 mark]

Correct amount of DNA as shown by thickness of band or shade of band. [1 mark]

OR

ORE:

- Model of DNA replication: Dispersive replication [1 mark]



Correct positioning of the bands. [1 mark]

Correct amount of DNA as shown by thickness of band or shade of band. [1 mark]

[L2]

Fig. 3.2 shows a simplified representation of DNA replication occurring on a linear chromosome.

The cellular process shown in Fig. 3.2 has many similarities with translation even though the products formed are different. Some of the similarities are that they both take place in 3 different stages (initiation, elongation, termination), both require energy, monomers for extension, a template for product synthesis as well as bond formation involving the removal of a water molecule.

(d) State three other similarities between the cellular process shown in Fig. 3.2 and translation.

.....

.....

.....

..... [3]

S: State (CW), cellular process...Fig. 3.2, translation, other similarities

C:

- Cellular process in Fig. 3.2: DNA replication
- Similarities between DNA replication & Translation
- Other similarities: Cannot mention both require energy, monomers for extension, a template for product synthesis as well as bond formation involving the removal of a water molecule

ORE:

1. Both processes require enzymes (DNA replication: DNA polymerase & Translation: Peptidyl transferase).
2. Both processes involve complementary base pairing.
3. Both processes are regulated by regulatory factors (DNA replication: cyclins, CDKs, Helicase & Translation: translational regulatory proteins).
4. Both processes are compartmentalised in the cell (DNA replication: nucleus, Translation: ribosome).
5. Errors can occur for both processes.
6. Both processes only occur when required by the cell.

[L3]

[Total: 10]

- 4 Fig. 4.1 shows two different stages, A and B (as shown by arrow) of the HIV reproductive cycle.

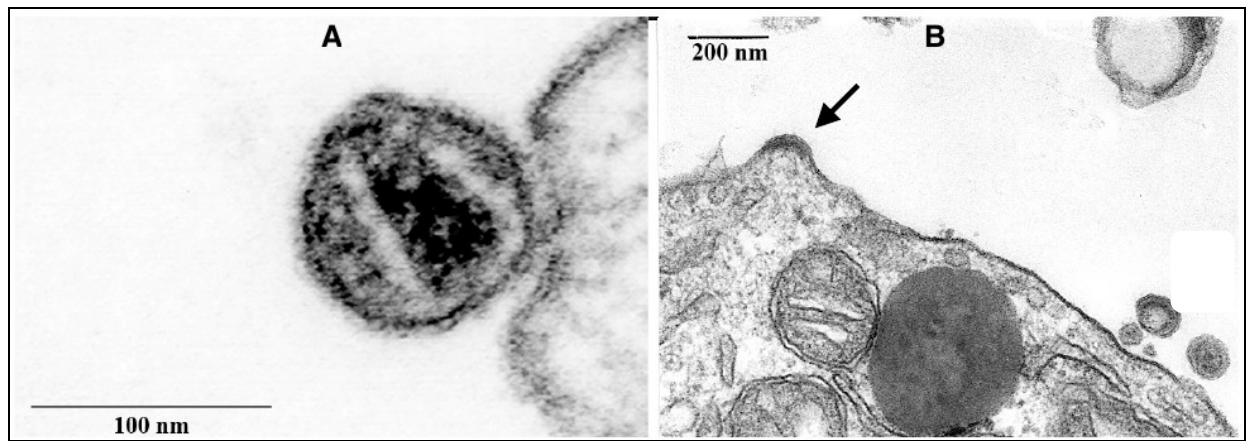


Fig. 4.1

- (a) Describe the events occurring in stage A of the HIV reproductive cycle.

.....
.....
.....[2]

S: Describe (CW), events, occurring, stage A, HIV reproductive cycle

C:

- Stage shown: Adsorption
- GP120 binds CD4 receptor and co-receptor CCR5/CXCR4.

ORE:

1. GP120 on viral envelope binds to CD4 receptor on host cell surface membrane.
2. GP120 also binds to co-receptor CCR5/CXCR4 on host cell surface membrane.

Reject: Fusion (as fusion is yet to occur as seen in Fig. 4.1)

[L2]

- (b) Compare the stage immediately following stage A with stage B.

.....
.....
.....[2]

S: Compare(CW), stage immediately following A, with B

C:

- Compare → Cite one difference and one similarity
- Stage immediately following stage A → Fusion
- Stage B → Budding
- Fusion vs Budding

ORE:

	Basis of comparison	Stage immediately following stage A (Fusion)	Stage B (Budding)
Difference	1. Virus entry/exit	Virus is <u>entering</u> the <u>host cell</u> during <u>fusion</u> .	Virus is <u>leaving</u> the <u>cell</u> via <u>budding</u> .
Difference	2. Membrane interaction	During fusion, <u>viral envelope fuses</u> with <u>cell surface membrane</u> . (Reject: Endocytosis)	During budding, virus <u>acquires host cell surface membrane</u> as <u>viral envelope</u> .
Similarity	3. Cytoskeleton involvement	Both processes involve <u>rearrangement</u> of the <u>cytoskeleton</u> at the cell surface membrane.	
Similarity	4. Location of capsid	For both processes, the viral <u>capsid</u> is not totally surrounded by the viral envelope and is partly <u>in</u> the host cell <u>cytoplasm</u> .	
Similarity	5. Requirement of energy	Both processes <u>require energy / ATP</u> .	
Similarity	6. Rearrangement of phospholipid	Both processes involve the <u>rearrangement of phospholipid</u> .	

(1 similarity and 1 difference, Max 2)

[L3]

(c) Contrast the final stage of the reproductive cycle of HIV and T4 phage.

.....
.....[1]

S: Contrast (CW), stage B, release stage, T4 phage

C:

- Contrast → Only cite difference (reject similarities)
- T4 phage undergoes lytic cycle
- Budding (of HIV) vs. Release via osmotic lysis (of T4 phage)

ORE:

1. Release of HIV via budding will not directly kill the host cell but release of T4 phages via osmotic lysis will kill the host cell.
2. For the release of T4 phages, a phage-encoded enzyme, lysozyme will break down the bacterial peptidoglycan causing osmotic lysis and release of the intact new bacteriophages whereas for the release of HIV no enzymes are involved.

(Max 1)

[L3]

For treatment, HIV-infected patients can receive HIV antiviral drugs such as Tenofovir (Fig. 4.2A).

Fig 4.1B shows an adenosine triphosphate (dATP) molecule, which shares similar chemical groups to Tenofovir.

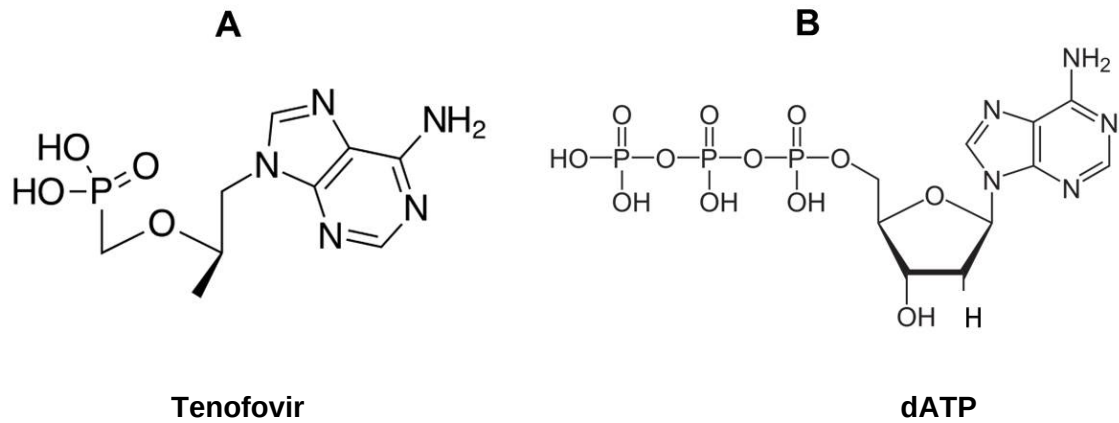


Fig. 4.2

- (d) With reference to Fig. 4.2, suggest how Tenofovir acts as a drug that interferes with the HIV reproductive cycle.

.....

[2]

S: Suggest how (CW), with reference to Fig. 4.2, Tenofovir, act..drug, interferes HIV cycle

C:

- With reference to Fig. 4.2 → Need to refer to structures of A & B
- A does not have 3' OH but B has.
- Enzyme competing for is reverse transcriptase
- A is competitive inhibitor
- But it causes chain termination → New incoming nucleotides cannot be added.

ORE:

Either 1 or 2

1. Tenofovir is an analog of adenosine triphosphate. / Tenofovir has a similar shape/conformation to adenosine monophosphate
OR
2. It is a competitive inhibitor for the reverse transcriptase enzyme. / It competes with adenosine triphosphate for the active site of reverse transcriptase.

Plus 3 or 4

3. which lacks an 3' OH group,
 - and hence results in chain termination (when incorporated into the existing DNA strand.
 OR
4. which lacks an 3' OH group,
 - / and incoming nucleotide cannot form phosphodiester bond with the DNA molecule.
5. Reverse transcription cannot be completed (OWTTE) resulting in a not complete/truncated viral DNA molecule to form.

(Max 2)

[L3]

HIV entry into cells requires involvement of at least one type of co-receptor. CCR5 is required for HIV virus entry. CCR5 Δ32 is a 32-base-pair deletion that introduces a premature stop codon into the CCR5 receptor locus, resulting in a non-functional receptor.

Timothy Ray Brown was an AIDS patient who received a hematopoietic stem cell transplant from a donor with homozygous CCR5 $\Delta 32$ on the CCR5 gene. After the transplant, he stopped his antiretroviral treatment. Following that, it was found that Timothy's HIV viral levels steadily decreased and his CD4 T- cell count increased. Eventually, he was found to be cured from HIV.

- (e) Suggest why not all HIV-infected patients can be cured with this therapeutic method.

.....
.....[1]

S: Suggest why (CWs), not all HIV-infected patients, cured, this therapeutic method

C:

- There are different strains of HIV
- Some strains use a different co-receptor for adsorption and then entry (e.g. CXCR4)

ORE:

1. There are different strains of HIV which use a different co-receptor, such as CXCR4 for adsorption and entry.

[L2]

- (f) Explain how HIV infection may result in the death of infected CD4 cells.

.....
.....
.....[2]

S: Suggest how (CWs), HIV infection, death, non-infected CD4 cells

C:

- HIV-infected cells release cytokines
- Syncytium form

ORE:

1. HIV-infected cells may undergo apoptosis.
2. Multiple bound cells may fuse, forming a giant multinucleated cell or syncytium. The syncytium may rupture / is destroyed by the body's immune system.

[L2]

[Total: 10]

5 Fig. 5.1 shows the formation of a eukaryotic transcription initiation complex.

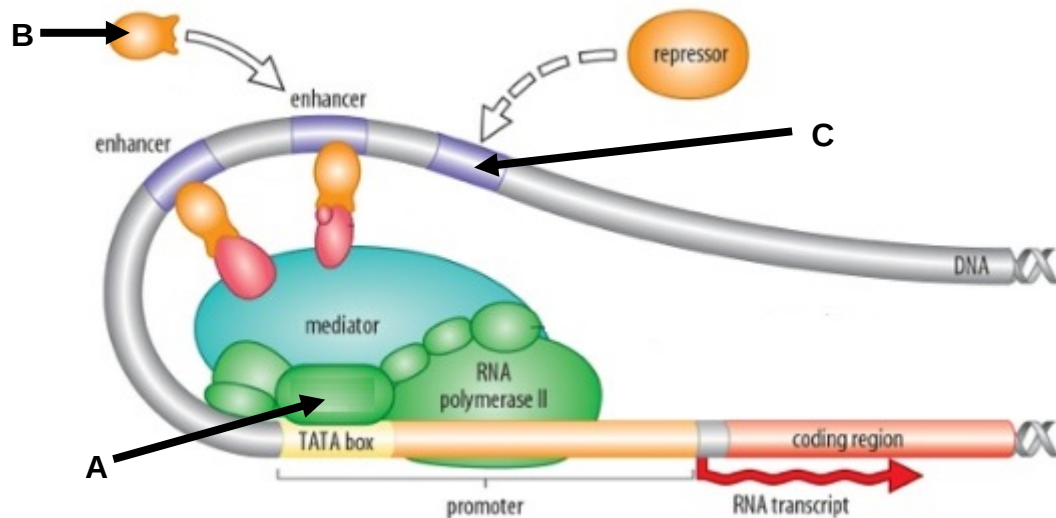


Fig. 5.1

(a) Identify the following proteins.

- A: _____ General transcription factor
- B: _____ Transcriptional activator protein
- C: _____ Silencer sequence

[3]

[L1]

(b) Explain how the eukaryotic transcription initiation complex for high rate of transcription can be formed.

-
-
-
-
-[4]

1. General transcription factors must assemble at the promoter to position RNA polymerase II correctly at the promoter and release it from the promoter into elongation mode for transcription.
 2. Binding to transcriptional activator proteins to enhancer,
 3. resulting in protein mediated bending of DNA / OWTTE
 4. to bring bound activators in contact with other proteins of the transcription initiation complex / Accept mediator protein action
 5. Stabilizing RNA polymerase for high rate of transcription.
- (Max 4)

[L2]

High rate of transcription caused by mutations in cancer critical genes may result in dysregulation of cell cycle control and subsequently lead to uncontrolled cell division.

c-myc is a regulator gene that codes for a transcription factor. A mutated version of c-myc that is highly expressed is found in many cancers. Fig 5.2 shows the relative level of expression of c-myc in normal and cancer prostate tissue.

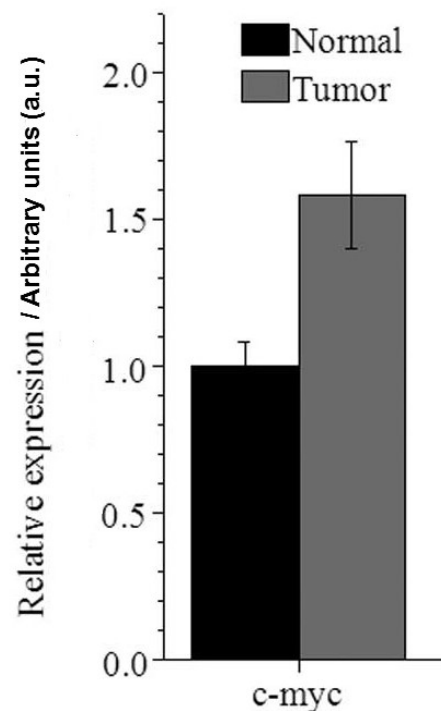


Fig 5.2

- (c) With reference to Fig.5.2, explain how mutation in the c-myc gene led to increased expression in cancer prostate tissue.

.....

[3]

1. c-myc expression in normal prostate tissue is 1.0 a.u. as compared to 1.6 a.u in cancer prostate tissue.
2. c-myc proto-oncogene undergoes gain-of-function mutation to become oncogene.
3. Translocation beside active promoter / Gene amplification resulted in increased expression of c-myc.

[L3]

[Total: 10]

- 6 Fig. 6.1 shows a Calico cat with a mosaic coat with patches of orange and black. It is known that fur coat colour in cats is determined by a single gene. Only female cats can develop calico fur coat. Male cats usually have only orange or black fur coat.

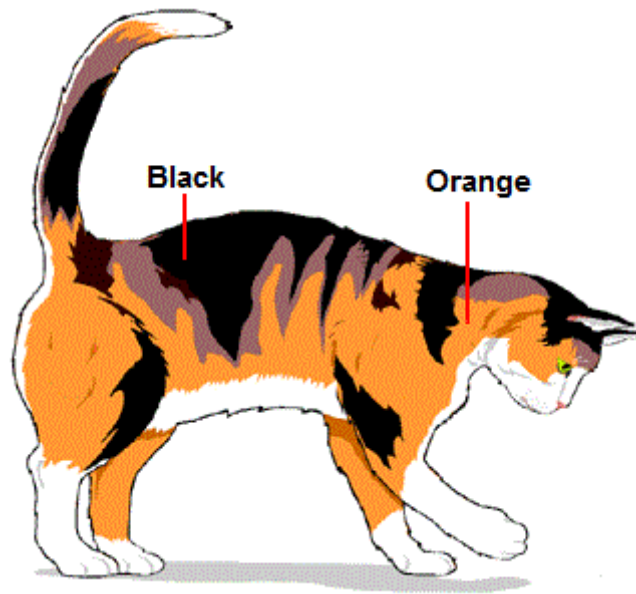


Fig. 6.1

- (a) Identify the type(s) of inheritance determining Calico fur coat colour in cats.

.....
.....[1]

S: State (CWs), type of inheritance, fur coat colour

C:

- Calico (patches of black and orange) → both equally expressed → codominance/epistasis.
- Difference between male and female → possible sex-linked.
- Inheritance determined by single gene → confirm codominance (cannot be epistasis)

ORE:

1. Co-dominance
2. Sex-linked / X-linked

[L2]

- (b) Using B to represent allele for black coat and R to represent allele for orange coat, draw a genetic diagram to show how a cat-breeder can obtain Calico cat from a cross between a pure-breeding black male and an orange female cat.

S: Draw (CWs), genetic diagram, orange female x black male.

C:

- Sex-linked & codominant (X^B & X^R alleles)
- Orange female $X^R X^R$
- Black male $X^B Y$

ORE:

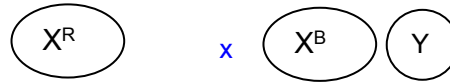
Let X^B represent the allele for black coat colour.

Let X^R represent the allele for orange coat colour, where X^B and X^R are codominant.

Parental phenotypes : Orange Female x Black Male

Parental genotypes (2n): $X^R X^R$ x $X^B Y$

Parental gametes (n):



F₁ genotypes and phenotypes (2n):

	♂	X^B	Y
♀	X^R	$X^B X^R$ Calico coat	$X^R Y$ Orange coat

F₁ phenotypic ratio: 1 Calico female: 1 Orange male

[L2]

[4]

Coat colour inheritance in horses is different from cats. Two unlinked genes *E* and *G* control coloured coat in horses. The two genes are thought to be involved in the same metabolic pathway for pigment formation.

- Horses may be bay, black or chestnut in colour.
- Horses may be bay / black when at least one dominant allele *E* is present.
- Chestnut coat colour is always produced in the presence of two copies of the *e* allele.

Horses coat colour goes through a natural graying process. Horses born with bay, black or chestnut coat colour will steadily turn gray. This process is mediated by a single copy of the dominant allele *G* regardless of the genotype of the gene *E* controlling coat colour.

- (c) Draw a genetic diagram in the space below to show the result of the cross between two gray horses that were heterozygous at both gene loci *G/g* and *E/e* and the resultant phenotypic ratio of the offspring.

S: Draw (CWs), genetic diagram, *GgEe* (Gray) x *GgEe* (Gray) Dominant epistasis

C:

- Epistatic gene locus *G* (result in gray coat regardless of genotype of gene *E*)
- Hypostatic gene locus *E*
- Single copy of *E* allele will result in horses with black/bay coat
- Homozygous recessive (*ee*) will result in horses with chestnut coat

ORE:

Let *G* represent the dominant allele for Gray coat colour and *g* represent the recessive allele for other colour coat.

Let E represent the dominant allele for black/bay coat colour, where e represent the recessive allele for chestnut coat.

Gene locus G is epistatic over Gene locus E

[1]

P phenotypes : Gray Horse x Gray Horse

[1]

P genotypes (2n) : GgEe x GgEe

P gametes (n): GE Ge gE ge x GE Ge gE ge

[1]

Punnett square:

♂ gametes ♀ gametes	GE	Ge	gE	ge
GE	GGEE Gray	GGEe Gray	GgEE Gray	GgEe Gray
Ge	GGEe Gray	GGee Gray	GgEe Gray	Ggee Gray
gE	GgEE Gray	GgEe Gray	ggEE Black/Bay	ggEe Black/Bay
ge	GgEe Gray	Ggee Gray	ggEe Black/Bay	ggee Chestnut

[1]

Offspring phenotypic ratio : 12 Gray: 3 Black/Bay: 1 Chestnut

[1]

[L3]

[5]

[Total: 10]

7 Fig. 7.1 shows the different modes of signal conduction in a non-myelinated neuron.

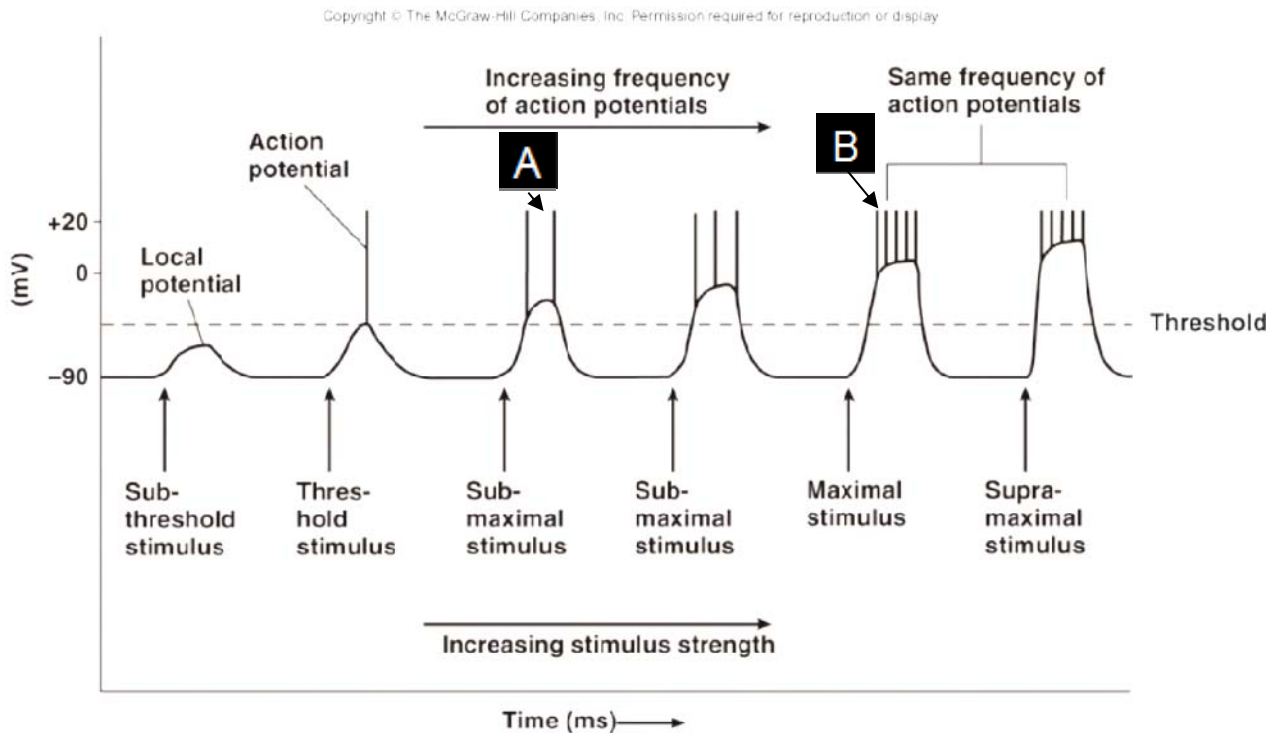


Fig. 7.1

(a) State one factor that contributes to unidirectional conduction of impulses down an axon.

.....[1]

SC: one factor / contributes / unidirectional conduction / neuron.

OR: refractory period

1. Refractory period (accept absolute and relative refractory periods)

[L1]

With reference to Fig. 7.1 for all questions following,

(b) explain why there is no action potential generated with a sub-threshold stimulus.

.....

[2]

SC: no AP / sub threshold stimulus.

OR: 1. Sub-threshold stimulus does not reach threshold.

2. No opening of VG Na⁺ Channels, no influx of Na⁺ and subsequently no AP created. .

1. Sub-threshold stimulus does not reach threshold therefore no opening of all voltage-gated Na⁺ Channels.

2. no influx of Na⁺ and subsequently no AP created.

[L2]

- (c) explain how the time interval between action potentials seen in label **B** is shorter than the time interval between action potentials seen in label **A**.

.....
.....
.....
.....[3]

SC: time interval / B / shorter than / A.

OR: 1. Refractory period
2. dependent on strength of signal
3. B maximal stimulus - A sub maximal stimulus.

1. Time interval between action potentials is dependent on the refractory period
2. and on the strength of the incoming signal.
3. B has maximal stimulus compared to A which has sub maximal stimulus.

[L3]

- (d) state the advantages in the ability of neurons to display these characteristics shown in Fig. 7.1.

.....
.....
.....[2]

SC: advantage / neuron ability / display characteristics .

OR: 1. Variation / differentiation of signal frequency.
3. determines the sensitivity of the neuron / OWTTE.

[L2]

- (e) explain what would increase the speed of the signal conduction down the neuron.

.....
.....
.....[2]

SC: increase speed/ down neuron.

OR: 1. Myelin sheath
2. Saltatory conduction

1. With the presence of the myelin sheath
2. Signal conduction down the neuron is even faster due to saltatory conduction.
3. at the nodes of Ranvier.

(Max 2)

[L2]

[Total: 10]

- 10 The Grand Canyon National Park is home to two groups of squirrels. The Albert squirrels *Sciurus aberti* live generally on the south rim of the canyon and the Kaibab squirrels *Sciurus kaibabensis* live on the north rim of the canyon (Fig. 8.1 and Fig. 8.2).

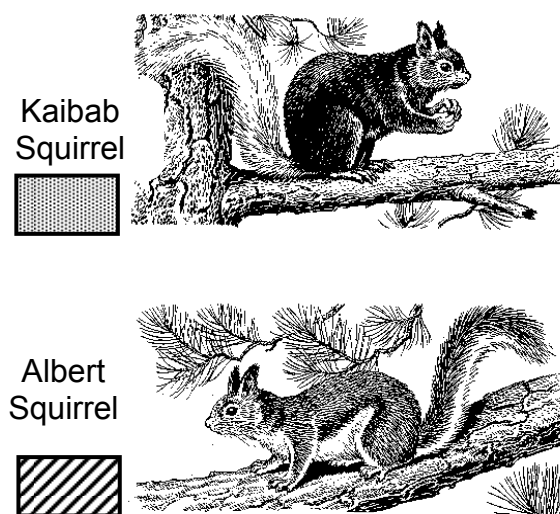


Fig. 8.1

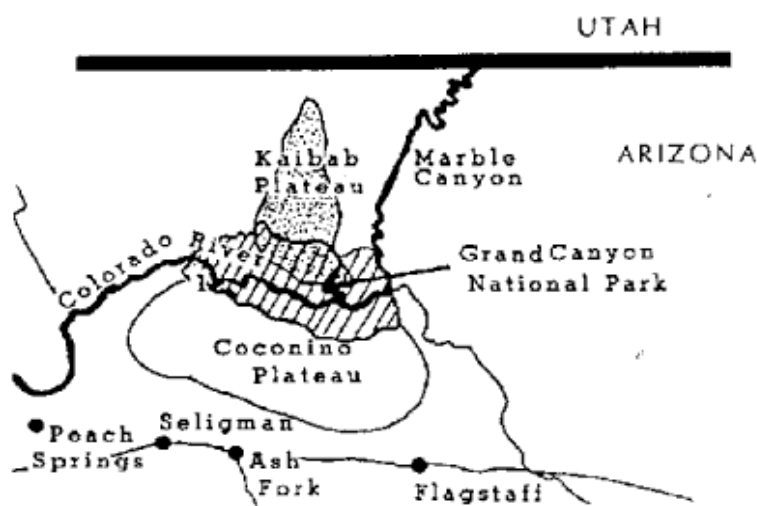


Fig. 8.2

The north rim is about 370 m higher than the south rim. Almost twice as much precipitation falls on the north rim than on the south rim every year. The two groups share many characteristics, but they do not look the same, both groups have tasselled ears, but each group has a unique fur colour pattern.

- (a) Explain which Darwinian principles can be applied from the above information on *S. aberti* and *S. kaibabensis*.

.....

.....

.....

.....

.....[4]

SC: which Darwin Principles / squirrels

OR: 1. High reproductive potential- great range

2. Competition / struggle to survive – no information

3. Constant population- great range seen in both groups.

4. Variation- Seen from the different phenotypic characteristics.

5. Selection pressure – environment on either side of canyon.precipitation

6. Passed down to next generation- phenotypical characteristics.

1. High reproductive potential inferred from the wide distribution of both groups in Fig. 8.2
2. Constant population inferred from the wide distribution of both groups in Fig. 8.2
3. Variation can be inferred from the different phenotypic characteristics of the squirrels.
4. Selection Pressure inferred from the environmental conditions on either side of the canyon, reference to precipitation being twice greater in the north compared to the south.
5. Passed down to the next generation- phenotypical characteristics of each group.

[L2]

Several studies have been done on the phylogenetic relationship of the squirrels in and around the Grand Canyon region. Fig. 8.3 is one such study based on cytochrome b DNA sequences.

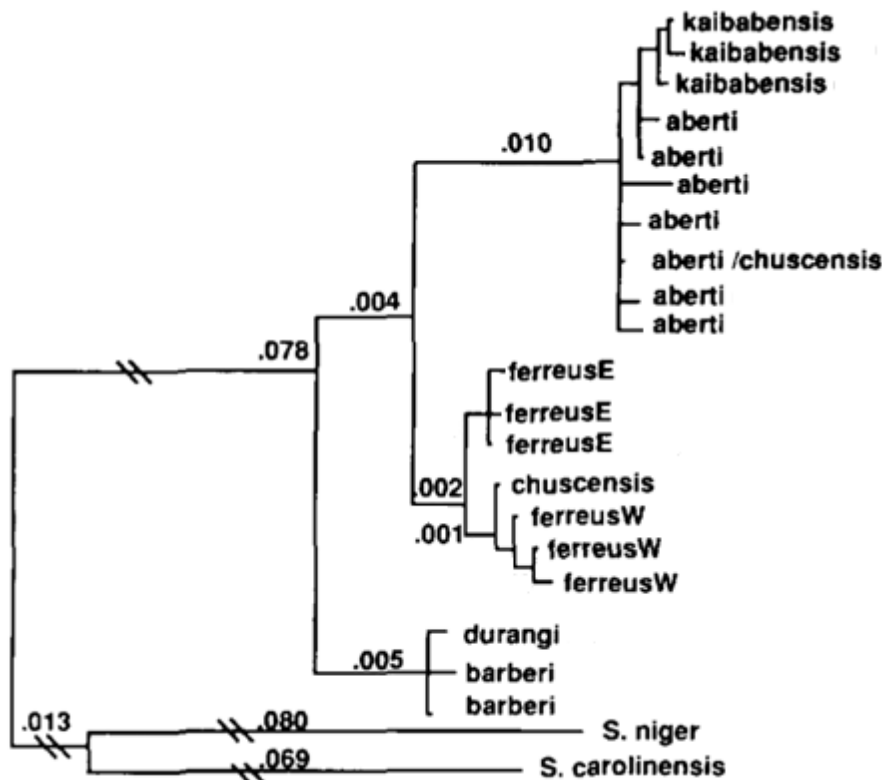


Fig.8.3 Phylogenetic relationship between six *sciurus* subspecies base on cytochrome b sequences constructed by the neighbour-joining method of Saitou and Nei (1987) using the *S. niger* and *S. carolinensis* (Thomas and Martin 1993) sequences as outgroups. Branch lengths and confidence probabilities are noted above and below the branches respectively.

- (b) With reference to information already given and also to Fig. 8.3, it is clear that divergent evolution or adaptive radiation is occurring in the evolution of *S. alberti* and *S. kaibabensis*

Explain why it is not convergent evolution.

.....

[2]

SC: kind of evo- not divergent / adaptive rad. / why not convergent evo

OR: 1. common ancestor

2. subspecies

Reject morphological homology

1. share a recent common ancestor

2. convergent evolution involves two phylogenetically different groups with no recent common ancestor.

Or any one of the following-

3. homology of DNA sequence for cytochrome b shows high relation 0.01

4. they are subspecies

(Max 2)

[L2]

- (c) (i) With reference to information already given and also to Fig. 8.3, suggest with reasons what kind of speciation *S. Alberti* is undergoing.

.....

.....

.....[2]

SC: kind of speciation

OR: 1. Probably allopatric speciation.

2. Due to the presence of a physical barrier (a river).

OR

3. Probably sympatric speciation.

4. Due to presence of overlapping geographical regions.

(Points 1 and 2 are awarded together, points 3 and 4 are awarded together)

[L2]

- (ii) With a known species concept, explain what would be the determining factor confirming *S. alberti* and *S. kaibabensis* as two separate species.

.....

.....

.....[2]

SC: determining factor / confirming separate species

OR: 1. Random mating between *S. alberti* and *S. kaibabensis* to confirm no fertile offspring
2. Reproductive isolation

1. Probably no fertile offspring can be derived between the two species.

2. Confirming reproductive isolation as a definitive indication via biological species concept.

3. Phylogenetic - Molecular evidence (DNA + RNA or amino acid sequence comparison).
Reject: Ecological (insufficient information), Morphological (insufficient information from just comparing phenotype).

[L3]

[Total: 10]

Section B

Answer **one** question. Answer each part on a **separate** piece of paper.

Write your answers on separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 9 (a) Describe cell signaling with the G-protein coupled receptor with a named ligand and its corresponding cellular response. [8]

1. The ligand, glucagon binds to the G-protein coupled receptor (GPCR). This causes a conformational change in the GPCR.
2. G-protein will be activated and undergo a conformational change causing GDP on G-protein will be substituted for GTP.
3. Alpha subunit of the G-protein will translocate to membrane-bound enzyme, adenylyl cyclase and activate so that it catalyse the conversion of ATP to cAMP.
4. cAMP will act as a second messenger and activate protein kinase A.
5. Protein kinase A will catalyse the phosphorylation of another protein kinase resulting in its subsequent activation.
6. A phosphorylation cascade will occur.
7. Eventually the a specific hydrolytic enzyme / glycogen phosphorylase will be activated,
8. which will hydrolyse glycogen to glucose monomers (glucose-1-phosphate).

[L1]

- (b) Explain the advantages and disadvantages of a phosphorylation cascade. [6]

Advantages	Great degree of control – can be stopped at any stage.	1
	Sensitive- can be reset very easily via phosphatases.	2
	Acts as a buffer before the final cellular response.	3
	Is triggered by 2 nd messengers which affect more than one metabolic process.	4
	<u>Amplification</u>	5
Disadvantage	Energy consuming ATP needed for each activation of PKA / OWTTE	6
	Will be susceptible to temperature/pH etc.	7
	A mutation that renders a protein kinase non-functional may terminate the cascade.	8
	A mutation that renders a phosphatase non-functional may cause the cascade to continue.	9

[L3]

- (c) Contrast with elaboration, anaerobic respiration with light independent reaction (Calvin cycle).[6]

	Anaerobic respiration	Light independent reaction
Reduced co-enzymes	1. Uses <u>NADH</u> in the	2. Use of <u>NADPH</u> in the conversion
Reduction process	3. <u>reduction of pyruvate to ethanol</u> .	4. conversion of <u>glycerate-3 phosphate</u> to <u>glyceraldehyde-3 phosphate</u> .
ATP	5. <u>ATP synthesized</u> via <u>substrate level phosphorylation</u> .	6. <u>ATP used</u> in the <u>conversion of glycerate 3 phosphate to glyceraldehyde 3 phosphate</u>
CO ₂	7. <u>CO₂ produced</u> in the <u>conversion of pyruvate to ethanol / lactic</u> .	8. <u>CO₂ incorporated</u> with <u>RuBP</u> to <u>form citric acid</u> .

	<u>acid</u>	
Location	9. Occurs in <u>cytoplasm</u> .	10. Occurs in the <u>stroma</u> .
Molecule regenerated	10. <u>NAD⁺</u>	11. <u>RuBP</u> & <u>NADP⁺</u>

[L3]

[Total: 20]

10 (a) Contrast binary fission and mitosis.

[6]

S: Contrast (CW), binary fission, mitosis

C:

- Contrast → Cite differences
- Binary fission vs mitosis (must have clear basis of comparison)
 - Type of cell that process occurs in
 - Location of cell that process occurs in
 - etc.....

ORE:

Basis of comparison	Binary fission	Mitosis
Type of cells that process occurs in	1. Binary fission occurs in <u>prokaryotes</u> .	1. Mitosis occurs in <u>eukaryotes</u> .
Location of cell that process occurs in	2. Binary fission occurs in the <u>nucleoid / cytoplasmic</u> region of prokaryotic cell / OWTTE	2. Mitosis occurs in the <u>nucleus</u> of eukaryotic cell./ OWTTE
Type of DNA involved	3. Binary fission occurs on <u>circular</u> double-stranded <u>DNA</u> .	3. Mitosis occurs on <u>linear</u> double stranded <u>DNA</u> .
Number of chromosomes involved	4. Only <u>1 chromosome</u> is involved in binary fission.	4. <u>More than 1 chromosome</u> is involved in mitosis.
Separation of daughter cells	5. There is <u>division</u> of <u>parental cell</u> to give rise to <u>2 daughter cells</u> during binary fission.	5. There is <u>no division</u> of <u>parental cell</u> to give rise to 2 daughter cells during mitosis. This occurs during cytokinesis.
Type of division	6. Involves <u>cell division</u> .	6. Involves <u>nuclear division</u> .
Origin of replication	7. DNA replication occurs at <u>1 point of origin</u> .	7. DNA occurs at <u>multiple points</u> of the genome.
Utilisation of spindle fibres	8. There is <u>no spindle fibre formation</u> during binary fission.	8. <u>Spindle fibre formation</u> occurs during mitosis.
Presence/absence of end replication problem	9. There is <u>no End replication problem</u> during binary fission.	9. <u>End replication problem occurs</u> during mitosis.

Reject: Citation of interphase and cytokinesis as these stages occur prior and after mitosis.

[L3]

(b) Describe generalised and specialised transduction.

[8]

S: Describe (CW), specialised transduction

C:

- Content knowledge recall

ORE:

Generalised transduction:

1. A virulent phage infects a bacteria cell and hydrolytic enzymes degrade the host bacteria chromosome into fragments.
 2. A small fragment of the host cell's degraded DNA is improperly packaged within a capsid,
 3. rather than the phage genome due to an error during the viral particle assembly process.
 4. When this phage attaches/infects to another bacteria cell, it will inject this foreign bacterial DNA into its new host.
 5. and may be integrated into new host bacteria chromosome under the proper circumstances
 6. via homologous recombination.
- (Max 4)

Specialised transduction:

7. A temperate phage infects a bacteria cell and has its viral DNA genome integrated into the bacterial chromosome.
 8. When the prophage viral DNA is excised from the chromosome,
 9. it sometimes takes with it a small region of adjacent bacterial DNA due to improper excision.
 10. These bacterial DNA are injected, along with the phage's genome, into the next host bacteria cell and
 11. may be integrated into new host bacteria chromosome under the proper circumstances.
 12. via homologous recombination.
- (Max 4)

(Max 4 marks for generalised, Max 4 marks for specialised)

[L1]

(c) In an experiment using T4 bacteriophage, different component molecules were labelled.

- T4 bacteriophages with protein coats labelled with radioactive sulfur.
- T4 bacteriophages with DNA labelled with radioactive phosphorus.

The differently-labelled bacteriophages were allowed to infect host bacteria.

The inside and outside of infected bacteria before lysis were tested for radioactive sulfur and radioactive phosphorus.

Describe and explain the expected results of the experiment.

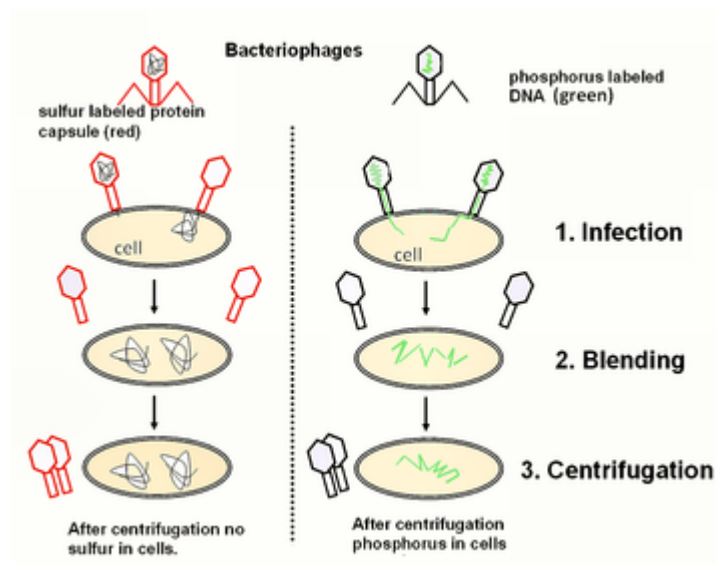
[6]

S: Describe..Explain (CWs), expected results, Protein coats...labelled..radioactive sulphur, DNA..labelled..radioactive phosphorus, inside...outside..bacteria...before lysis...tested for radioactive

C:

- Description
 - Bacterial cells infected with radioactive sulfur-labelled bacteriophages:
 - Radioactivity will be detected outside

- No radioactivity will be detected inside
- Explanation
 - Protein capsid does not enter the bacteriophage, hence only outside will have radioactivity
- Description
 - Bacterial cells infected with radioactive phosphorus-labelled bacteriophages:
 - No radioactivity will be detected outside
 - Radioactivity will be detected inside
- Explanation
 - DNA will be injected into the bacterial cell during transduction, hence DNA that is radioactive phosphorus-labelled will be detected inside the bacterial cells.



ORE:

Bacteria infected with radioactive sulfur-labelled bacteriophages:

Description:

1. There will be radioactivity detected outside the bacteria.
2. There will be no radioactivity detected inside the bacteria.

Explanation:

3. Protein capsids do not enter the bacteria cells during transduction / but instead remain on the bacteria cell surface membrane after transduction.

Bacteria infected with radioactive phosphorus-labelled bacteriophages:

Description:

4. There will be radioactivity detected outside the bacteria.
5. There will be no radioactivity detected inside the bacteria.

Explanation:

6. Bacteriophage DNA enter /is injected into the bacteria cells during transduction.

[Total: 20]

END OF PAPER